E-commerce Checkout Drop-off Analysis

Final Project Story & Interview Explanation

60-90 Second Project Explanation

I built an end-to-end e-commerce cart abandonment analysis using PostgreSQL and Power BI. The dataset contained event-level customer activity such as product views, cart additions, and purchases, covering about 9.24 million sessions.

I first used SQL to validate and transform the event data into session-level metrics. One important issue I discovered was that some sessions had purchase events without recorded cart events, which initially caused the cart-to-purchase rate to exceed 100%. I corrected the funnel logic by separating Viewed Only, Cart Abandoned, Cart-to-Purchase, and Purchase Without Cart sessions.

After correcting the funnel, I found that qualifying cart sessions had approximately a 50.05% abandonment rate.

I then analyzed behavioral patterns associated with abandonment. Single-item carts had 53.72% abandonment, sessions with 11 or more product views reached 61.29%, and sessions with 6 or more repeat views had 59.11% abandonment. Abandoned sessions were also longer and involved more product exploration than converted sessions.

Finally, I created a Power BI dashboard with an executive overview, abandonment-driver analysis, and category-opportunity analysis. I prioritized categories using both abandonment and business volume rather than simply ranking by abandonment percentage.

Based on the analysis, I recommended testing checkout simplification for single-item carts, decision-support features for high-exploration shoppers, and targeted improvements in high-priority product categories. Since the data is observational, I framed these as hypotheses to validate through A/B testing rather than claiming causation.

If Asked: What Was the Most Challenging Part?

The biggest analytical challenge was defining the funnel correctly. Initially, purchase sessions exceeded cart sessions because the dataset contained purchases without recorded cart events. Instead of treating every purchase as a cart conversion, I investigated the event sequence and separated Purchase Without Cart sessions from true Cart-to-Purchase sessions. That changed the metric from a misleading conversion rate above 100% to a valid cart conversion rate of 49.95%.

If Asked: What Was Your Most Important Insight?

One of the strongest behavioral patterns was that abandonment increased substantially among customers doing extensive product exploration. Sessions with 11 or more product views had 61.29% abandonment, and sessions with 6 or more repeat views had 59.11%. Abandoned sessions also lasted longer and contained more product views than converted sessions. I interpreted this as a possible decision-friction signal and recommended testing comparison tools and clearer product information.

If Asked: What Business Impact Would This Analysis Have?

The main value is prioritization. Instead of applying the same checkout solution everywhere, the business can identify high-friction customer behaviors and categories where abandonment coincides with meaningful volume. That allows product and CRO teams to focus experiments where there is greater potential impact and measure the result using conversion, abandonment, and recovered revenue.

If Asked: Why Power BI?

SQL handled the heavy transformation and analytical logic, while Power BI provided an interactive layer for stakeholders. I structured the dashboard into three questions: what is happening, why might it be happening, and where should the business act first.

If Asked: What Would You Do Next?

My next step would be experimentation. For example, because high product exploration is associated with abandonment, I would test a product-comparison feature specifically for high-exploration sessions. I would measure incremental cart conversion against a control group. I would also want additional dimensions such as device, traffic source, shipping cost, payment failures, and customer type to determine whether they explain some of the observed patterns.

Key Numbers to Remember

Metric	Result
Total Sessions	9.24M
Cart Abandonment Rate	50.05%
Cart-to-Purchase Rate	49.95%
Single-item Cart Abandonment	53.72%
11+ Product Views Abandonment	61.29%
6+ Repeat Views Abandonment	59.11%
Avg. Abandoned Session Duration	14.42 min
Avg. Converted Cart Session Duration	10.53 min

One-line Resume / LinkedIn Version

Analyzed 9.24M e-commerce sessions using PostgreSQL and Power BI to identify cart-abandonment drivers, correct funnel-tracking anomalies, segment high-friction customer behavior, and prioritize product categories for conversion optimization.

Interview Reminder

Focus on the business problem, the funnel-definition challenge, two or three strong findings, and the actions you recommended. Do not try to explain every SQL query or dashboard visual unless the interviewer asks. Describe behavioral findings as associations and present recommendations as hypotheses to validate through controlled experiments.